Access INFN websites through an SSH SOCKS proxy

Linux · macOS · Windows 10/11 | Google Chrome and Mozilla Firefox

This setup routes websites under infn.it through zoroastro.to.infn.it, while other browsing stays on your normal connection. You need an SSH account on that server.

1 Start the SSH tunnel

Open Terminal (Linux/macOS) or PowerShell / Windows Terminal (Windows) and run:

```
ssh -N -D 127.0.0.1:1080 USERNAME@zoroastro.to.infn.it
```

Replace USERNAME with your INFN SSH login, authenticate, and leave the terminal open. Stop the tunnel with Ctrl+C. Port 1080 on your computer is now a SOCKS5 proxy.

Windows note: Try ssh -V first. If the command is missing, install OpenSSH Client under Windows Settings → Optional features (the exact menu path varies by Windows version). No PuTTY is needed.

2 Install FoxyProxy Standard

Install the same browser extension in either browser: Chrome Web Store | Firefox Add-ons.

3 Add the proxy in FoxyProxy

Click the FoxyProxy toolbar icon → Options → Proxies → Add. Enter:

Setting	Value
Title	INFN SSH
Proxy type	SOCKS5
Hostname	127.0.0.1
Port	1080
Proxy username/password	Leave blank (SSH handles authentication)

In Firefox, enable Proxy DNS / remote DNS for the SOCKS5 proxy if the option is shown, especially for internal INFN hostnames. Chrome normally resolves SOCKS5 destinations through the proxy.

4 Route only infn.it traffic

In the proxy's Patterns section, choose Add Pattern and configure Include with type Match Pattern (not Wildcard). Enter:

```
*://*.infn.it/*
```

Save, then select Proxy by Patterns in the FoxyProxy toolbar menu. The pattern covers HTTP/HTTPS requests for infn.it and its subdomains; nonmatching sites connect directly.

5 Test and troubleshoot

Open an INFN-only website. If it does not load, check that (1) the SSH terminal is still connected, (2) FoxyProxy is set to Proxy by Patterns, and (3) the site hostname matches the pattern. External sign-in pages or resources may require additional patterns.

Security: SSH encrypts traffic from your machine to zoroastro. HTTPS should still be used for destination websites. The SSH server must permit TCP forwarding and be able to reach the requested site.